

Get to Know Cloud Simulations Using FOSS

This article delves into cloud simulations, which is a fast-evolving domain. Virtual machines, hypervisors, data centres and cloud simulation tools like CloudSim and GreenCloud are covered extensively.

You must be aware that in cloud computing, remote resources are used on demand and there is no physical infrastructure at the client end. The actual resources are installed and deployed at remote locations, and are accessed remotely by different network-based protocols. There is a provision for computational resources on-demand, with end users being unaware about the physical location and configuration of the servers.

Virtualisation technology and the cloud

Virtualisation is the major technology that works with cloud computing. Dynamic virtual machines are created to provide end users and developers access to actual infrastructure at another remote location.

A virtual machine or VM is the software implementation of any computing device, machine or computer that executes the programs as a physical (actual) machine. When a user or developer works on a virtual machine, all the resources including programs installed on the remote machine are accessible using a specific set of protocols. For the end user of the cloud service, the virtual machine acts like the actual machine.

The term VM was originally proposed and defined by Popek and Goldberg as "...an efficient, isolated duplicate of a real machine."

Virtual machines are classified into two groups,

depending on their use and degree of correspondence to any real machine:

- **System virtual machine:** A system virtual machine provides a complete system platform that supports the execution of a complete operating system (OS). It emulates an existing system architecture and is built for two purposes. The first is to provide a platform to run programs where the real hardware is not available. The second is to have multiple instances of VMs leading to more efficient use of computing resources, in terms of energy consumption and cost effectiveness, or both. This is known as hardware virtualisation, which is the key to a cloud computing environment.
- **Process virtual machine (language virtual machine):** This type of VM is designed to execute a single program, which means that it supports a single process. Such VMs are closely suited to one or more programming language and are built to provide program portability and flexibility. An essential characteristic of a VM machine is that the software running inside is limited to the resources and abstractions provided by the VM - it cannot fragment or break out of its virtual environment.

Prominent virtualisation software

Here's a list of the major Windows- and Linux-based virtualisation software: